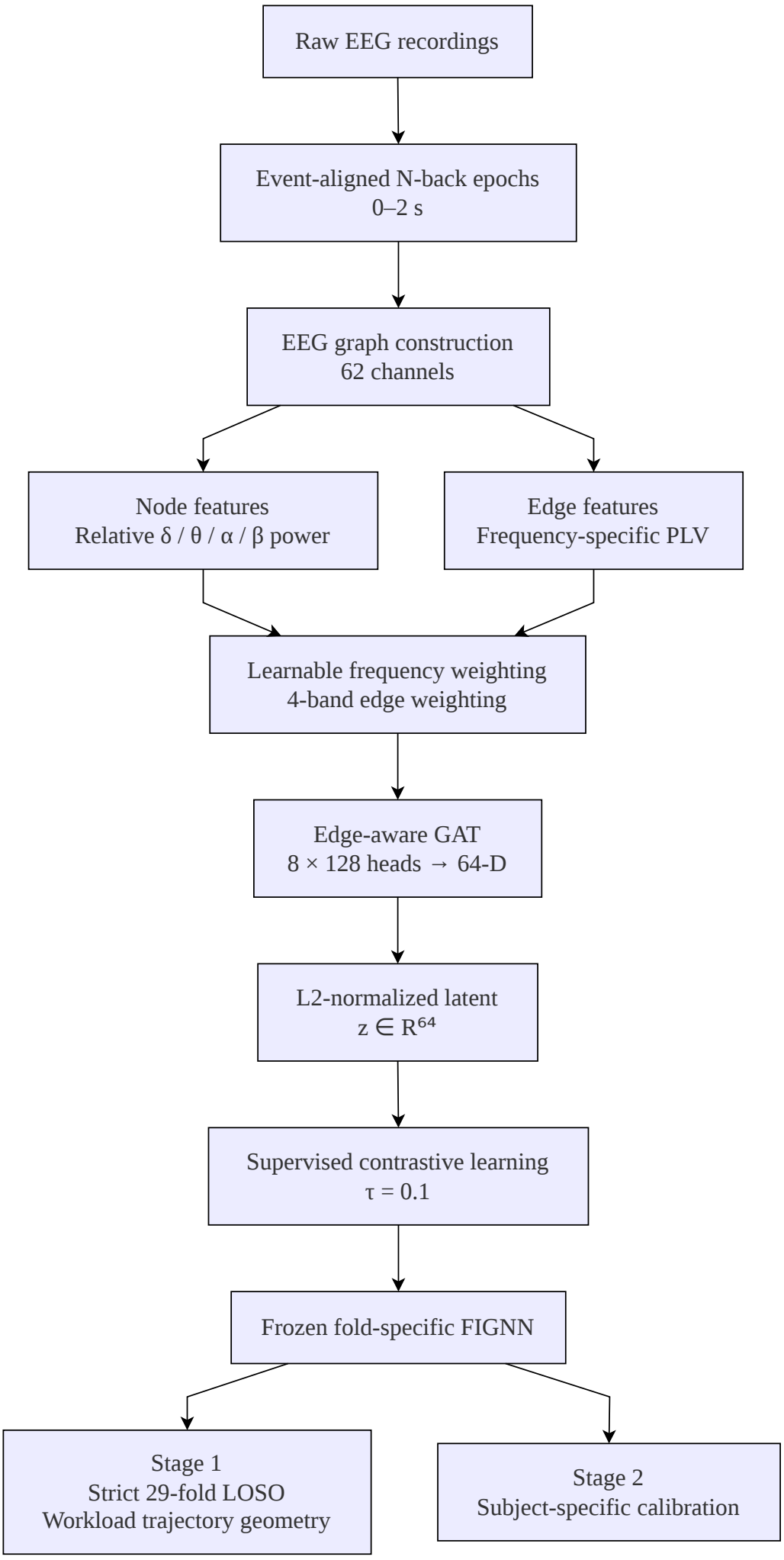
Raw EEG recordings
Event-aligned N-back epochs
0–2 s
EEG graph construction
62 channels
Node features
Relative δ / θ / α / β power
Edge features
Frequency-specific PLV
Learnable frequency weighting
4-band edge weighting
Edge-aware GAT
8 × 128 heads → 64-D
L2-normalized latent
z ∈ R⁶⁴
Supervised contrastive learning
τ = 0.1
Frozen fold-specific FIGNN
Stage 1
Strict 29-fold LOSO
Workload trajectory geometry
Stage 2
Subject-specific calibration